Companion Spectral Extraction via Joint Template Fitting in the Whitened Residual Space

Broadband Speckle Imaging Pipeline — Technical Note v2

---

1. Noise Model

Observed counts are Poisson with mean equal to the expected signal:

$$d_l(x) \sim \mathrm{Poisson}(m_l(x))$$

Variance (Poisson + read noise):

$$\sigma_l^2(x) = m_l(x)/g + \sigma_{\mathrm{read}}^2$$

Subtracting the satellite model removes the satellite *signal* but not its shot noise. **Use the noise-free satellite model for the variance** (not the data) to avoid a systematic bias in the regression — see Section 7.

2. Forward Model

The data in frame l is a broadband image — sum over wavelength slices k:

$$m_l(x) = \sum_k [o_k \star h_{k,l}](x)$$

There is no per-slice data frame. Spectral information is encoded in the wavelength-dependence of $h_{k,l}$ (speckle decorrelation).

3. Object Decomposition

Object at slice k = satellite s_k plus companion of amplitude α_k at position x_0:

$$o_k(x) = s_k(x) + \alpha_k\, \delta(x - x_0)$$

Model separates into satellite term M_l and companion term C_l:

$$M_l(x) = \sum_k [s_k \star h_{k,l}](x)$$

$$C_l(x) = \sum_k \alpha_k\, h_{k,l}(x - x_0)$$

4. Satellite Subtraction and Residual

After reconstruction converges (outside-support pixels zeroed):

$$r_l(x) = d_l(x) - \hat{M}_l(x) = \sum_k \alpha_k h_{k,l}(x - x_0) + n_l(x)$$

The residual contains companion signal from **all wavelength slices simultaneously** through their respective PSFs.

5. Irreducible Noise Floor

$$\sigma_l^2(x) \approx \hat{M}_l(x)/g + \sigma_{\text{read}}^2$$

This is set by the satellite shot noise halo at the companion position and cannot be reduced by better reconstruction. The matched filter is the only path below it (coherent summation across frames).

6. Joint Spectral Regression

Fit all α_k simultaneously to the broadband residual:

$$\mathcal{L}_C = \sum_l \sum_x \frac{[r_l(x) - \sum_k \alpha_k h_{k,l}(x - x_0)]^2}{\hat{M}_l(x)/g + \sigma_{\text{read}}^2}$$

This is linear in all α_k. Setting all partial derivatives to zero gives the 10×10 linear system $\mathbf{A}\alpha = \mathbf{b}$:

$$A_{kk'} = \sum_{l,x} \frac{h_{k,l}(x - x_0)\, h_{k'l}(x - x_0)}{\sigma_l^2(x)}$$

$$b_k = \sum_{l,x} \frac{r_l(x)\, h_{k,l}(x - x_0)}{\sigma_l^2(x)}$$

$$\hat{\boldsymbol{\alpha}} = \mathbf{A}^{-1}\mathbf{b}$$

7. Critical: Model-Based Variance (not data-based)

Using σ^2 = data/g + σ_{read}^2 introduces a systematic negative bias: when data fluctuates low, r_l is negative *and* inv-σ^2 is high, so negative residuals are up-weighted. This produces a large negative b even when the companion signal is positive.

Fix: use the noise-free satellite model for the variance ($\sigma^2 = M\blacksquare_l/g + \sigma_{read}^2$). This breaks the correlation between weights and residual noise. Validated: noise-free self-test gives ratio=1.000 at all slices.

8. PSF Template Construction

Template $T_{kl}(x) = h_{kl}(x–x_0)$ via FFT shift theorem, matching the sign convention used in data generation:

$$T_{kl}(x) = \mathrm{Re}\left[\mathcal{F}^{-1}\left\{\mathrm{FTpsf}_{kl} \cdot e^{-i2\pi(r_0 f_y + c_0 f_x)}\right\}\right]$$

Only aperture pixels within companion_ap_rad of x_0 are stored (T_ap[k,l,n_ap]). Aperture radius ~160 px captures >96% of PSF volume (FWHM ~70 px).

9. Cross-Slice Overlap Matrix A

- **Diagonal A_{kk}** = Fisher information per slice. Predicted $SNR_k = \alpha_k\sqrt{A_{kk}}$.
- **Off-diagonal $A_{kk'}$** = spectral cross-talk from PSF similarity (all PSFs from same scaled OPD). Condition number ~18.
- **A^{-1}** = spectral covariance matrix. Its diagonal gives $\sigma(\alpha\blacksquare_k)$; off-diagonals give spectral correlations between slice estimates.

10. Uncertainties, SNR and Connection to Matched Filter

$$\sigma(\hat{\alpha}_k) = \sqrt{[A^{-1}]_{kk}}, \quad \mathrm{SNR}_k = \hat{\alpha}_k/\sigma(\hat{\alpha}_k)$$

b_k is the matched filter output at x_0 for slice k. The broadband MF coadd = $\sum_k b_k$. The joint regression is a spectrally-decomposed, cross-talk-corrected MF.

11. Fidelity Metrics

- **Pearson r**: shape fidelity (1=perfect).
- **Spectral angle** = $\arccos(\alpha\blacksquare \cdot \alpha_{truth} / |\alpha\blacksquare||\alpha_{truth}|)$ degrees (0=perfect).
- **RMSFE** = rms fractional error (0=perfect amplitude).
- **Amplitude ratio** = $\mathrm{mean}(\alpha\blacksquare_k/\alpha_k)$ (1=unbiased; reconstruction imperfection gives ~1.5x).

12. Validated Results (mv=11.9, Δm=5.9, 0.25″, 100 frames)

- Noise-free / truth model: ratio=1.000 at all 10 slices.
- Noisy / truth model: SNR 4–10 per slice. Pearson r=0.83, spectral angle=9.6°.
- Noisy / reconstructed model: SNR 7–14 per slice with ~1.5x amplitude bias.
- Direct reconstruction: Pearson r=0.48, spectral angle=26.3°, amp ratio=0.51.

Inputs: restart=1, niter=0, support_mask_file set, companion_spectrum=1, companion_ds9_x/y from MF detection, companion_ap_rad~160px. Outputs: companion_spectrum.txt, companion_spectrum_covariance.txt, companion_spectrum.png.